

## **Palantir Foundry Enablement Program for RIT Team**

**Target Audience:** Developers, Technical Leads, QA Engineers, Architects, Delivery Team Members

**Duration:** 40 Hours

**Schedule:** 4 Weekends (Saturday & Sunday)

**Session Duration:** 5 Hours Per Day

**Mode:** Instructor-Led with Hands-on Exercises and Case Study

---

### **Training Objectives**

By the end of this program, the RIT team will be able to:

- Understand the Palantir Foundry platform architecture and ecosystem
- Work with Ontology Objects and operational workflows
- Build applications using Workshop and Slate
- Create and manage data pipelines
- Develop Code Repositories and Functions
- Use Pipeline Builder and Contour effectively
- Understand Foundry administration and governance basics
- Participate in enterprise Palantir delivery projects
- Build use-case driven solutions aligned to rail/logistics operations

### **Training Methodology**

- Concept Explanation
- Guided Demonstrations
- Hands-on Labs
- Mini Assignments
- End-to-End Case Study
- Architecture Discussions

- Delivery Best Practices
- 

## **Weekend 1 — Foundations & Core Platform**

### **Day 1 — Palantir Fundamentals and Platform Overview (5 Hours)**

#### **Introduction to Palantir Ecosystem**

- Introduction to Palantir
- Foundry vs Gotham vs AIP
- Enterprise Use Cases
- Palantir in Transportation & Rail Industry
- Data Operating System Concept

#### **Foundry Platform Architecture**

- Foundry Core Components
- Data Lineage
- Object-Centric Architecture
- Dev/Prod Lifecycle
- Workspace Navigation

#### **Foundry User Experience**

- Home Workspace
- Projects and Resources
- Catalog
- Marketplace
- Ontology Overview
- Data Security Concepts

#### **Hands-on Labs**

- Foundry Navigation
- Project Creation

- Dataset Exploration
- Lineage Analysis
- Marketplace Exploration

### **Case Study Introduction**

- Rail Operations Use Case
- Shipment Tracking Scenario
- Train Movement Monitoring
- Asset Visibility Concepts

## **Day 2 — Data Integration & Pipeline Foundations (5 Hours)**

### **Data Connection Concepts**

- Structured vs Unstructured Data
- Data Sources
- Batch vs Streaming
- Data Integration Patterns

### **Pipeline Builder**

- Pipeline Builder Overview
- Data Transformations
- Joins and Aggregations
- Data Quality Checks
- Scheduling Basics

### **Code Repositories Basics**

- Repository Structure
- Branching Concepts
- Python and SQL in Foundry
- Incremental Processing

## **Hands-on Labs**

- Build First Pipeline
- Create Transformations
- Data Cleaning Exercises
- Incremental Pipeline Example

## **Best Practices**

- Pipeline Design Standards
- Naming Standards
- Reusability
- Performance Considerations

## **Weekend 2 — Ontology & Application Development**

### **Day 3 — Ontology Fundamentals (5 Hours)**

#### **Ontology Introduction**

- What is Ontology?
- Object-Centric Modeling
- Object Types
- Links and Relationships
- Actions and Functions

#### **Ontology Manager**

- Creating Object Types
- Defining Properties
- Relationships and Link Types
- Ontology Security

#### **Operational Use Cases**

- Shipment Object Model

- Rail Asset Modeling
- Maintenance Workflows
- Exception Management

### **Hands-on Labs**

- Create Ontology Objects
- Create Relationships
- Configure Actions
- Object Exploration

### **Ontology Best Practices**

- Modeling Guidelines
- Reusability Patterns
- Scalability Design
- Governance Concepts

## **Day 4 — Workshop & Application Building (5 Hours)**

### **Workshop Fundamentals**

- Workshop Overview
- Operational Applications
- Layouts and Widgets
- Variables and Interactions

### **Building Operational Apps**

- Object Tables
- Metrics and Charts
- Filters and Drilldowns
- Action Workflows

### **Application Design**

- UX Best Practices
- Role-Based Views
- Operational Dashboards
- Exception Monitoring

### **Hands-on Labs**

- Build Shipment Dashboard
- Create Operational Views
- Configure User Actions
- Build KPI Monitoring Screen

### **Application Lifecycle**

- Publishing Applications
- Sharing and Permissions
- Versioning
- Deployment Basics

## **Weekend 3 — Advanced Development & Automation**

### **Day 5 — Functions, Logic & Automation (5 Hours)**

#### **Foundry Functions**

- Functions Overview
- Function Types
- Input/Output Handling
- Integration Patterns

#### **TypeScript Functions**

- Function Development
- Business Logic Implementation
- Error Handling

- Testing Functions

### **Actions & Automation**

- Action Types
- Workflow Automation
- Notifications
- Trigger Mechanisms

### **Hands-on Labs**

- Create Business Function
- Build Approval Action
- Automate Workflow Logic
- Trigger-Based Processing

### **Enterprise Design Practices**

- Modular Design
- Logging
- Monitoring
- Reusable Components

## **Day 6 — Contour, Analytics & Monitoring (5 Hours)**

### **Contour Overview**

- Analytical Exploration
- Data Visualization
- Business Insights

### **Advanced Analytics**

- Operational KPI Analysis
- Rail Movement Analytics
- Delay and Exception Analysis

## **Monitoring & Observability**

- Pipeline Monitoring
- Build Status Tracking
- Data Health Concepts
- Alerting Mechanisms

## **Hands-on Labs**

- Create Analytical Views
- Build Operational Metrics
- Configure Alerts
- Analyze Dataset Health

## **Optimization Techniques**

- Performance Tuning
- Cost Optimization
- Query Optimization
- Build Optimization

## **Weekend 4 — Enterprise Delivery & Capstone**

### **Day 7 — Governance, Security & Enterprise Delivery (5 Hours)**

#### **Foundry Administration Overview**

- User Management
- Roles and Permissions
- Access Controls
- Security Model

#### **Enterprise Governance**

- SDLC in Foundry
- Dev/Test/Prod Promotion



- Release Management
- Audit and Compliance

### **Delivery Methodology**

- Palantir Project Lifecycle
- Agile Delivery Model
- Team Structure
- Environment Management

### **QA & Testing Strategies**

- Testing Pipelines
- Ontology Testing
- Application Testing
- Regression Strategies

### **Delivery Best Practices**

- COE Standards
- Reusable Accelerators
- Documentation Practices
- Delivery Governance

## **Day 8 — End-to-End Capstone & Project Simulation (5 Hours)**

### **Capstone Use Case Overview**

- Rail Logistics Scenario
- Operational Requirements
- Business Objectives

### **Team-Based Implementation**

Participants will:

- Create Data Pipelines

- Build Ontology
- Configure Actions
- Develop Workshop Application

### **Application Demonstration**

- Team Presentations
- Architecture Walkthrough
- Design Decisions

### **Review & Optimization**

- Improvement Areas
- Scaling Considerations
- Production Readiness

### **Final Wrap-Up**

- Key Takeaways
- Recommended Learning Path
- Delivery Readiness Discussion
- Q&A Session

### **Deliverables Included**

- Training Material
- Hands-on Lab Exercises
- Sample Datasets
- End-to-End Case Study
- Architecture Reference Materials
- Best Practices Guide
- Capstone Exercise

**Recommended Audience**

- Developers
- Technical Architects
- QA Engineers
- Delivery Leads
- Data Engineers
- Business Analysts
- Integration Engineers

**Expected Outcomes**

After completion of this program, the RIT team will be capable of:

- Participating in Palantir implementation projects
- Building operational applications
- Working with Ontology-driven solutions
- Supporting enterprise delivery activities
- Developing pipelines and automation
- Contributing to rail/logistics transformation initiatives